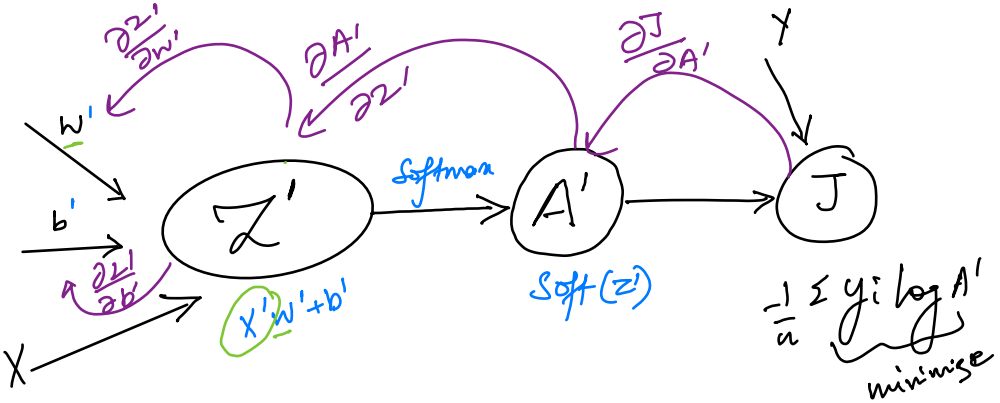
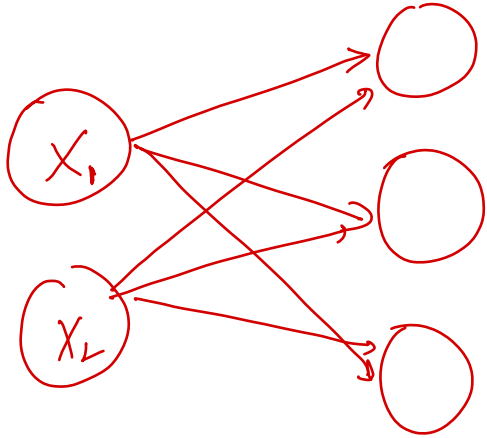
Agenda :

- Activation function
- Vanishing Gradient
- Notation
- Forward Propogation





2 features
I/P layer

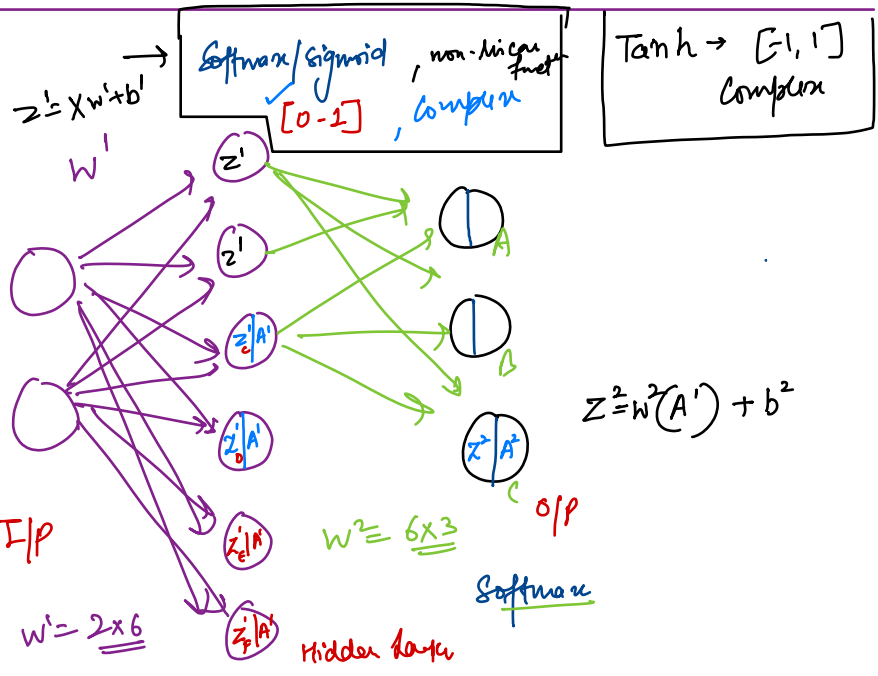
3 class in O/P layer

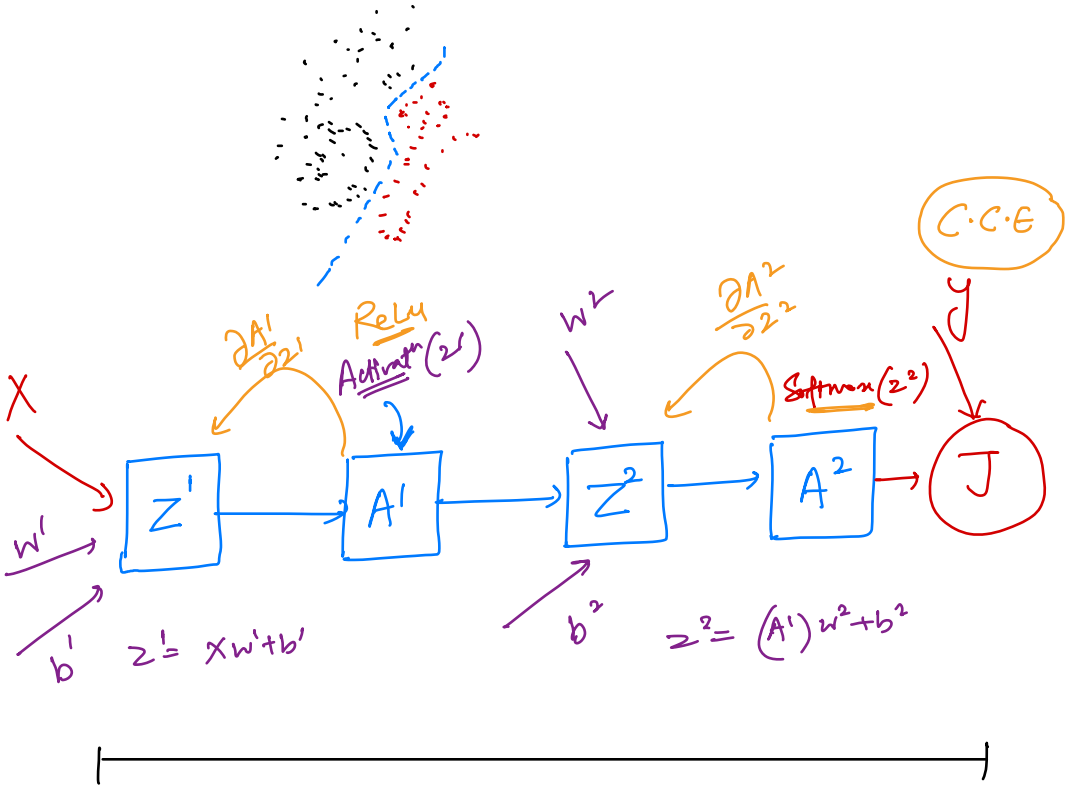
$$\frac{\partial J}{\partial w^1} = \frac{\partial J}{\partial A^1} \times \frac{\partial A^1}{\partial z^1} \times \frac{\partial z^1}{\partial w^1} \Rightarrow x$$

$$\frac{\partial J}{\partial b^1} = \underbrace{\frac{\partial J}{\partial A^1} \times \frac{\partial A^1}{\partial z^1}}_{a-y} \times \frac{\partial z^1}{\partial b^1}$$

$$\frac{\partial J}{\partial w^1} = (a-y)\, x$$

$$\frac{\partial J}{\partial b^1} = (a-y) \cdot 1$$





Imagine there are multiple hidden layers

$\tanh(A^1) \longrightarrow \cdots \tanh(A^2) \cdots \longrightarrow \tanh(A^3) \cdots \longrightarrow \tan(A^4) \cdots$

$$\frac{\partial J}{\partial W^1} = \frac{\partial J^1}{\partial A^4} \times \frac{\partial A^4}{\partial Z^4} \times \frac{\partial Z^4}{\partial A^3} \times \frac{\partial A^3}{\partial Z^2} \times \frac{\partial Z^2}{\partial A^1} \times \frac{\partial A^1}{\partial Z^1} \cdots$$

≈ 0

tanh : [-1, 1]

$$\frac{\partial\left(\frac{1}{1+e^{z}}\right)}{\partial z} \qquad \frac{\partial\left(\frac{1}{1+e^{z}}\right)}{\partial z}$$

$\frac{\partial \tanh}{\partial z} \rightarrow [0, 0.5, 1]$ (0.5)

$[0, 0.5, 1]$ (0.5)

$\frac{\partial \sin x}{\partial x} \rightarrow [0, 0.25]$

0.9 .... 0.9 ... 0.9

$\Rightarrow 0.081 \times 0.9$

$\simeq 0$

Vanishing Gradient problem → Sigmoid & tanh

Exploding Gradient problem

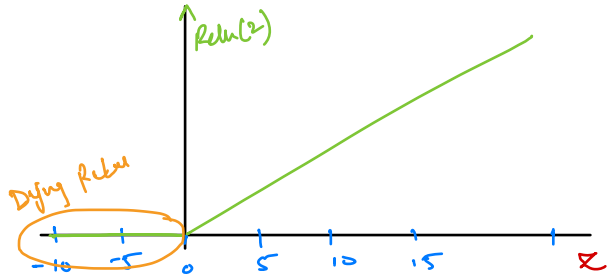
→ Non-linear
→ not Complex

ReLU →

$z \rightarrow \max(z, 0)$

$z = -50$ Relu(z) → 0

$z = 100$ Relu(z) → 100



$$\rightarrow \text{Relu}(z) = \begin{cases} 0 & z \le 0 \\ z & z > 0 \end{cases}$$

$$\rightarrow \text{Relu}'(z) = \begin{cases} 0 & z \le 0 \\ 1 & z > 0 \end{cases}$$

One Small issue

$z \le 0 \quad \text{Relu}(z) = 0$

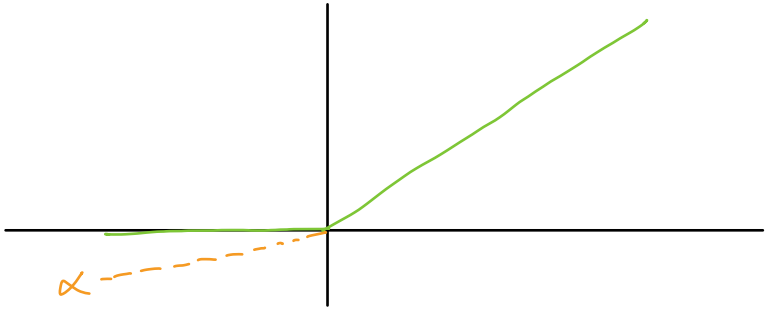
$z \le 0 \quad \text{Relu}'(z) = 0$

All -ve values, $\frac{\partial J}{\partial A} = 0$

→ no learning at all

potential vanishing gradient problem.

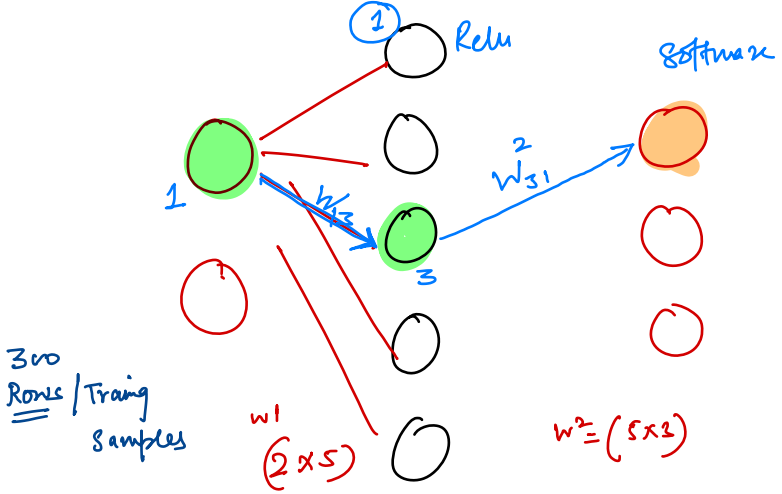
Leaky Relu



$z \le 0$

$(LR)' = \alpha$

$(LR)' = 1 \quad z > 0$



$X = A^0$ $\quad A^1$ $\quad A^2$

Z^1 $\quad Z^2$

Shape

$X = A^0$ $\quad (300, 2)$

X_1	X_2

$Z^1 = (300, 5)$ $\quad X \cdot W + b$

Relu(Z) $A^1 = (300, 5)$

$w^1 = (2x5)$

$b^1 = (1x5)$

$Z^2 = (300 \times 3)$

$A^2 = (300 \times 3)$

$w^2 = (5x3)$

$b^2 = (1x3)$

A	B	C
0	0	1
1	0	0

$(300, 3)$

$Z^1 = (300, 2) \times (2x5) + (1x5)$

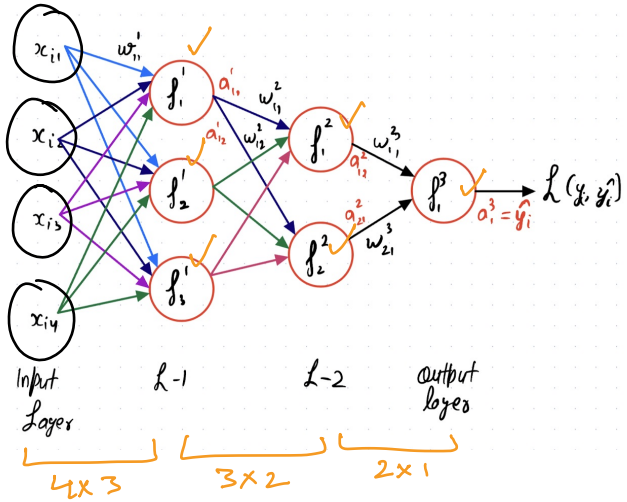
$\Rightarrow (300, 5) + (1x5)$ (broadcasting)

$Z^1 = (300, 5)$

$Z^2 = (A^1) w^2 + b^2$

$\Rightarrow (300, 5) \cdot (5, 3) + (1x3)$

$= (300 \times 3) + (1x3) \Rightarrow (300 \times 3)$



W= 12 6 2 $\Rightarrow$ 20

biases = 3 2 1 = 6

$\Rightarrow$ 26